SPETRO GEOPORTAL ACADEMY | OFFICIAL COURSE HANDBOOK

Cloud-Based Remote Sensing with Google Earth Engine

Author: David Saah, Nicholas Clinton et al.

Scope: Practical Field Manual & Reference Guide | Pages: 14

Course Overview & Objectives

The fundamental open-access handbook for planetary-scale Earth observation, time-series analysis, land-use classification, and water resources monitoring.

MODULE OUTLINE:

- **1. Earth Engine Architecture & Data Model**
- **2. Satellite Imagery Collections**
- **3. Spectral Indices & Cloud Masking**
- **4. Land Cover Classification & Machine Learning**
- **5. Hydrology & Environmental Change Detection**

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1. Earth Engine Architecture & Data Model

Client vs. server execution model, ee.Image, ee.ImageCollection, ee.Feature, ee.FeatureCollection, reducers, and pyramid tile caching.

KEY TAKEAWAYS & PRACTICAL LAB INSTRUCTIONS

1. Load corresponding dataset layers from the SPETRO Data Harvester or Hugging Face.
2. Execute geoprocessing algorithms and verify outputs in the Interactive Map Editor.
3. Export cartographic layouts, GeoTIFF rasters, or GeoJSON vector polygons for field use.

2. Satellite Imagery Collections

Working with Sentinel-2 MSI (Level-2A BOA reflectance), Landsat 8/9 OLI/TIRS, MODIS daily products, and Copernicus DEM at planetary scale.

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3. Spectral Indices & Cloud Masking

QA60 and SCL-based cloud and cirrus masking. Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), and Enhanced Vegetation Index (EVI).

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4. Land Cover Classification & Machine Learning

Random Forest, CART, and SVM classifiers in Earth Engine. Sample collection, training data partitioning, confusion matrix validation, and kappa statistics.

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5. Hydrology & Environmental Change Detection

Bi-temporal change detection, JRC global surface water dynamics, flood extent mapping using Sentinel-1 SAR dual-polarization (VV/VH), and drought index monitoring.

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